

ERROR DETECTION AND MEASUREMENT VALIDATION ARCHITECTURE

In various embodiments, the intraluminal compliance and distance profiling device incorporates error detection and measurement validation mechanisms configured to ensure that recorded measurements represent valid physiologic conditions. Validation systems may operate through hardware sensing, computational analysis, procedural workflows, or combinations thereof.

Measurement validation may include detection of motion artifacts caused by user movement, device instability, or environmental disturbance. In embodiments incorporating inertial sensing, acceleration and orientation data may be analyzed to determine whether motion exceeds predefined stability thresholds. Measurements obtained during unstable conditions may be rejected, repeated, or flagged with reduced confidence scores.

The system may further evaluate signal quality characteristics associated with distance sensing modalities. Validation algorithms may monitor signal return strength, measurement variance, sampling consistency, or temporal stability to confirm reliable ranging performance. Measurements exhibiting abnormal variability or low signal confidence may be automatically discarded or recalculated.

For compliance characterization, validation may include analysis of pressure response continuity, inflation rate consistency, and expected deformation behavior. Non-physiologic pressure signatures, abrupt pressure discontinuities, or abnormal expansion patterns may indicate improper positioning, leakage, or incorrect usage conditions and may trigger automated safety responses or user guidance prompts.

In certain embodiments, the device implements physiologic plausibility modeling to confirm that measured values fall within expected biological ranges. Computational models may compare measurements against learned population distributions, historical user data, or internally derived baselines to detect implausible or spoofed inputs.

Validation architecture may further include sensor cross-checking in which multiple sensing modalities confirm measurement agreement. For example, pressure response, motion stability, and distance confidence may be evaluated jointly to determine overall session validity. Multi-factor validation improves robustness against erroneous or intentionally falsified measurements.

The device or associated software may assign a measurement confidence score representing the reliability of each recorded session. Confidence scoring may consider stability duration, sensor agreement, signal quality, calibration state, and user workflow completion. Only measurements exceeding predefined confidence thresholds may be stored as baseline profile data.

In certain embodiments, guided measurement workflows may enforce validation procedures through step-by-step user interaction. Software prompts may require confirmation of positioning, stabilization periods, or completion of predefined measurement stages before accepting results.

Error detection mechanisms may also monitor device health conditions including sensor malfunction, communication failure, abnormal power behavior, or component degradation. Fault detection routines may suspend operation, notify the user, or initiate safe shutdown procedures when unsafe or unreliable conditions are detected.

Accordingly, the disclosed validation architecture improves measurement reliability, prevents accumulation of invalid data, enhances longitudinal comparability, and strengthens trustworthiness of derived physiologic profiles generated by the device and associated computational systems.